

MADHUMITHA J

Software Engineer | Python & AI/ML

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SUMMARY

Computer Science undergraduate with a strong foundation in **software engineering, Python, data structures, algorithms, and AI/ML**, complemented by hands-on experience developing data-driven applications. Skilled in **problem solving, machine learning, application development, model evaluation, and technical implementation**, with experience transforming complex requirements into functional solutions. Demonstrated ability to build and deploy AI-powered projects involving **predictive analytics, computer vision, and intelligent risk detection**, with a strong interest in developing scalable and reliable software solutions.

SKILLS

TECHNICAL SKILLS

Programming: Python, SQL

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Problem Solving, Debugging, Software Development, SDLC

AI & Machine Learning: Machine Learning, Deep Learning, Predictive Modeling, Feature Engineering, Model Evaluation, Classification, Computer Vision, Explainable AI

Frameworks & Libraries: Scikit-learn, Pandas, NumPy, OpenCV, Matplotlib, Seaborn, Librosa, YOLO, SHAP, Joblib

Web & Application Development: Streamlit, Flask, REST APIs

Databases & Tools: MySQL, Git, GitHub, Jupyter Notebook, VS Code

PROJECTS

AI-Powered Credit Card Fraud Detection System

Mar 2025 – May 2025

Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook

- **Developed** a machine learning application to identify **fraudulent credit card transactions** using transaction-level data and classification techniques.
- **Implemented** data preprocessing, **feature engineering**, exploratory data analysis, and model training using **Python and Scikit-learn**.
- **Evaluated** classification performance to improve the identification of **suspicious transactions** and support data-driven fraud detection.
- **Optimized** the prediction workflow through systematic data preparation and model evaluation.

Deepfake Voice Detection System

Nov 2025 – 03/2026

Technologies: Python, Librosa, Scikit-learn, NumPy, Pandas, Streamlit, Joblib

- **Developed** an end-to-end **audio classification application** to distinguish authentic speech from **AI-generated/deepfake voices**.
- **Engineered** an audio-processing pipeline using **MFCC and spectrogram feature extraction** for effective voice representation.
- **Implemented** the complete **training, preprocessing, classification, and inference pipeline** using Python and Scikit-learn.
- **Integrated** the trained model with a **Streamlit application** and Joblib-based model serialization for interactive prediction.
- **Enhanced** model robustness through **model optimization and noise-handling techniques**.

XSafeNet - An Explainable Multimodal Predictive Risk Intelligence Framework for Vehicle Telemetry and Indian Road Hazards

Jun 2026 – Present

Technologies: Python, YOLO, OpenCV, Scikit-learn, Pandas, NumPy, SHAP, OpenWeather API, Streamlit

- **Engineered** a **multimodal AI software framework** integrating vehicle telemetry, road hazards, weather, and traffic data for **predictive risk assessment**.

- **Developed** a **YOLO-based computer vision pipeline** to detect and classify road hazards from driving videos.
- **Implemented** data-processing and **risk-fusion workflows** to generate dynamic risk scores from multiple contextual inputs.
- **Integrated** the **OpenWeather API** to incorporate real-time environmental data into the risk-analysis pipeline.
- **Applied SHAP-based Explainable AI** to interpret model predictions and identify key factors influencing risk decisions.
- **Built and deployed** an interactive **Streamlit dashboard** for visualizing hazard detections, risk levels, and predictive insights.

PROFESSIONAL EXPERIENCE

The Developers Arena - Data Science Intern

Jan 2026 – Mar 2026 | chennai,TamilNadu

- Conducted data cleaning, preprocessing, and exploratory data analysis (EDA) using Python for effective data preparation.
- Built a Customer Churn Analysis model to analyze customer behavior and support predictive business insights.

Nextgen Solutions- Python Intern

May 2025 – Jun 2025
Cuddalore, TamilNadu

- **Engineered** a Python-based job application form to **capture and validate candidate data** through structured input workflows.
- **Implemented** validation logic and structured data processing to **improve data integrity, accuracy, and application usability**.

EDUCATION

B.Tech Computer Science and Engineering

2023 – 2027 | Chennai,TamilNadu

Hindustan institute of Technology and Science

Current CGPA: 9.18/10

Higher Secondary Education

2022 – 2023 | chennai,TamilNadu

SRM Public School

CERTIFICATES

Senior Level Typewriting Certification (Distinction)

Government-Recognized Examination . high typing speed and accuracy with professional precision.

Artificial Intelligence - MyCaptain

Completed training in **Python, NumPy, Pandas, Data Structures, Machine Learning, Regression, Clustering, and AI fundamentals**, with hands-on experience in building and testing ML algorithms.

Generative AI – MyCaptain

Generative AI, Large Language Models (LLMs), Prompt Engineering, AI Applications

AI Ascend Program - Acquiring Data Level 1: AI Fundamentals – Accenture | FutureSkills Prime

Successfully completed training in **AI fundamentals, machine learning concepts, data analysis, and AI applications**.